

Lab 2:

Fundamental Mathematical Concepts Required for Machine Learning

Aim: To understand and implement the core mathematical concepts of linear algebra, calculus, probability and statistics that underpin Machine Learning algorithms.

Machine Learning models are, at their heart, mathematical functions whose parameters are learned from data, so a working knowledge of the underlying mathematics makes it far easier to understand why an algorithm behaves the way it does. Linear algebra (vectors, matrices, dot products, eigenvalues/eigenvectors) is used to represent data points as vectors and to describe transformations such as the ones performed inside PCA or a neural network layer. Calculus, particularly derivatives and gradients, is the basis of optimization: almost every model is trained by minimizing a loss function using Gradient Descent, which repeatedly moves the parameters in the direction opposite to the gradient of the loss. Probability and statistics — mean, variance, probability distributions, and Bayes' theorem — allow models to reason about uncertainty in the data and form the theoretical foundation of algorithms such as Naive Bayes and Logistic Regression. This lab implements each of these building blocks directly in Python so that later labs (Regression, Naive Bayes, PCA, etc.) can be understood as natural applications of these same core ideas.

Solved Example

Problem: A delivery drone needs to travel from point A(2,3) to point B(8,11) in a 2-D coordinate plane. Represent the displacement as a vector, and compute its magnitude (straight-line distance) and unit direction vector.

Approach: *The displacement vector is simply $B - A$. Its magnitude (Euclidean norm) gives the straight-line distance the drone must travel, and dividing the vector by its own magnitude gives a unit vector — a vector of length 1 that represents pure direction, independent of distance. This exact vector arithmetic (differences, norms, and normalization) underlies distance-based algorithms like KNN and K-Means used in later labs.*

Python Program:

```
import numpy as np

A = np.array([2, 3])
B = np.array([8, 11])

displacement = B - A
distance = np.linalg.norm(displacement)
unit_vector = displacement / distance

print("Displacement Vector:", displacement)
print("Distance (Magnitude):", distance)
print("Unit Direction Vector:", unit_vector)
```

Result / Discussion: *The displacement vector is [6, 8], and its magnitude works out to 10.0 (a clean 6-8-10 right triangle), so the unit vector is [0.6, 0.8]. This shows how any straight-line distance calculation used later for KNN or K-Means clustering is simply an application of the Euclidean norm computed here.*

Exercise Questions

Q1.

Scenario: *An e-commerce recommendation engine represents each customer's purchase pattern as a vector across product categories, and needs to know how 'similar' two customers are to recommend products.*

Task: Given two customer vectors representing spending across 4 categories, compute the cosine similarity between them (using dot product and norms) to quantify how similar their purchasing behaviour is.

Q2.

Scenario: *A factory wants to compress its sensor readings (originally correlated across 2 machines) into fewer independent directions of maximum variation.*

Task: Given a small 2-feature dataset, compute the covariance matrix and its eigenvalues/eigenvectors, and identify the direction of maximum variance.

Scenario: *A startup wants to fit the cheapest possible pricing function $y = mx + c$ to just 2 data points of cost vs. quantity manually, before trusting Scikit-learn's regression model used later in Lab 3.*

Task: Implement Gradient Descent from scratch to fit a line $y = mx + c$ to a small set of (x, y) points by minimizing the Mean Squared Error.

Q4.

Scenario: *A call center wants to know the probability that a randomly selected call will be answered within 30 seconds, given historical call-answering statistics that follow a known probability distribution.*

Task: Given that call-waiting times follow a Normal distribution with mean 40s and standard deviation 12s, use python package to compute the probability that a call is answered within 30 seconds.

Q5.

Scenario: *A spam-filtering startup (a preview of the Naive Bayes lab) wants to reason about the probability that an email is spam, given that it contains the word 'free', using known statistics about spam and non-spam emails.*

Task: Given $P(\text{Spam})=0.4$, $P(\text{'free'}|\text{Spam})=0.7$ and $P(\text{'free'}|\text{Not Spam})=0.1$, use Bayes' theorem to compute $P(\text{Spam} | \text{'free' appears in the email})$.

Q6.

Scenario: *A quality-control engineer measures the diameter of 12 manufactured bolts and wants a quick statistical summary to check if the production process is consistent (low variance) before approving the batch.*

Task: Compute the mean, variance, standard deviation, and coefficient of variation (std/mean) for the bolt diameters, and comment on consistency.

Q7.

Scenario: *A furniture manufacturer produces tables and chairs using wood and labour hours. Given the total wood and labour available, and the amount each product consumes, the production manager wants to know exactly how many tables and chairs to make to use up all resources exactly.*

Task: Represent the resource constraints as a system of linear equations (in matrix form $Ax = b$) and solve for the exact production quantities using matrix inversion.

Q8. *A startup is training a simple cost model that depends on two parameters (price and marketing spend) and wants to know the direction of steepest increase in cost at the current parameter setting, before applying gradient descent.*

Task: For the multivariable function $f(x, y) = x^2 + 3xy + y^2$, compute the partial derivatives df/dx and df/dy symbolically and evaluate the gradient vector at the point (2, 1).

Q9.

Scenario: A factory's quality-control team records, for 200 items, whether each item was produced on Machine A or Machine B and whether it was Defective or Not Defective, and wants to know the probability an item is defective specifically given that it came from Machine B.

Task: Given a contingency table of machine vs defect status, compute the conditional probability $P(\text{Defective} \mid \text{Machine B})$.

Q10.

Scenario: A marketing team wants hard statistical evidence (not just a scatter plot impression) of how strongly advertising spend and product sales are related, before deciding to increase the ad budget.

Task: Compute the Pearson correlation coefficient between weekly advertising spend and weekly sales, and interpret its strength and direction.

Q11.

Scenario: A call center manager knows that, historically, 20% of outbound calls result in a successful sale. Before staffing tomorrow's shift, she wants to know the probability that exactly 4 out of the next 10 calls will result in a sale.

Task: Model the number of successful sales out of 10 independent calls as a Binomial distribution with $p=0.2$, and compute the probability of exactly 4 successes, and the probability of at least 6 successes.